PAIC simulation results

2023-10-04

Overview of the problem

The goal is to estimate the relative effect of treatment A vs B, noted Δ_t. These two treatments have been evaluated in two distinct trials, AB and AC. The target estimand is the relative effect of A vs B in the source population of trial AC, written $\Delta_{t(AC)}$.

$\Delta_{t(AB)}$ will be estimated using different estimators:

- Unadjusted:
  - Unanchored: naive model
  - Anchored: difference of relative effects A versus C and B versus C
- Adjusted
  - Adjusted using a multivariable outcome model (regression model), or univariable weighted outcome model (propensity score)
    * Each of these models will be fitted using either Individual Patient Data (IPD) for both trials, or IPD for the trial AB and AgD for trial BC: Population-Adjusted Indirect Comparisons (PAIC). PAIC multivariable outcome model = Simulated Treatment Comparison (STC), univariable weighted outcome model = Matching-Adjusted Indirect Comparisons (MAIC).
    * Each of these models will be either anchored or unanchored.

(Current) scenario settings

- Sample size per treatment arm = 100
- One gaussian outcome Y
- Treatment effects:
  - $\beta_A = 1.5$
  - $\beta_B = 1$
  - $\beta_C = 0 (\textit{no effect, } \sim \textit{placebo})$
- One binary confounding factor X_1, following a binomial distribution $B(X_1) \sim 0.5$ in an overarching population from which patients of trials AB and AC have been sampled. Effects of this confounding factor on 1. trial assignment and 2. the outcome Y:
  1. Constant effect on the outcome (prognostic): $\beta_{Y_{X1}} = 1.5$, and effect on the outcome through interaction with treatment A (interaction): $\beta_{Y_{A:X1}} = 1.2$
  2. Effect on trial assignment ($\beta_{AC,X1} = 2$) and ($\beta_{BC,X1} = 0$)
- Outcome model :

$$Y \sim \left(\beta_{Y_{X_1:A}} A + \beta_{Y_{X1}}\right) X1 + \beta_{Y_A} A + \beta_{Y_B} B + \beta_{Y_C} C + \epsilon_Y$$

with $\epsilon_Y \sim N(0,1)$

- Trial assignment model : $P(Z = AC) \sim g(\beta_{AC,X1} X_1)$ with $g(X) = (1 + exp(-X))^{-1}$

```
prop_X1 <- 0.5
bT_X1 <- 2

bY_X1 <- 1.5
bYA_X1 <- 1.2

bY_A <- 1.5
bY_B <- 1.5
bY_C <- 0

N_RCT <- 200

trial_assignment_model_AC <- bquote(X1 * bT_X1)
trial_assignment_model_BC <- bquote(0)
outcome_model <- bquote(bYA_X1*X1*A + bY_X1*X1 + bY_A*A + bY_B*B + bY_C*C)
```

Monte-Carlo simulations settings

Two thousands iteration of that scenario, and estimation of variance of treatment effects via bootstrap (1000 iterations).

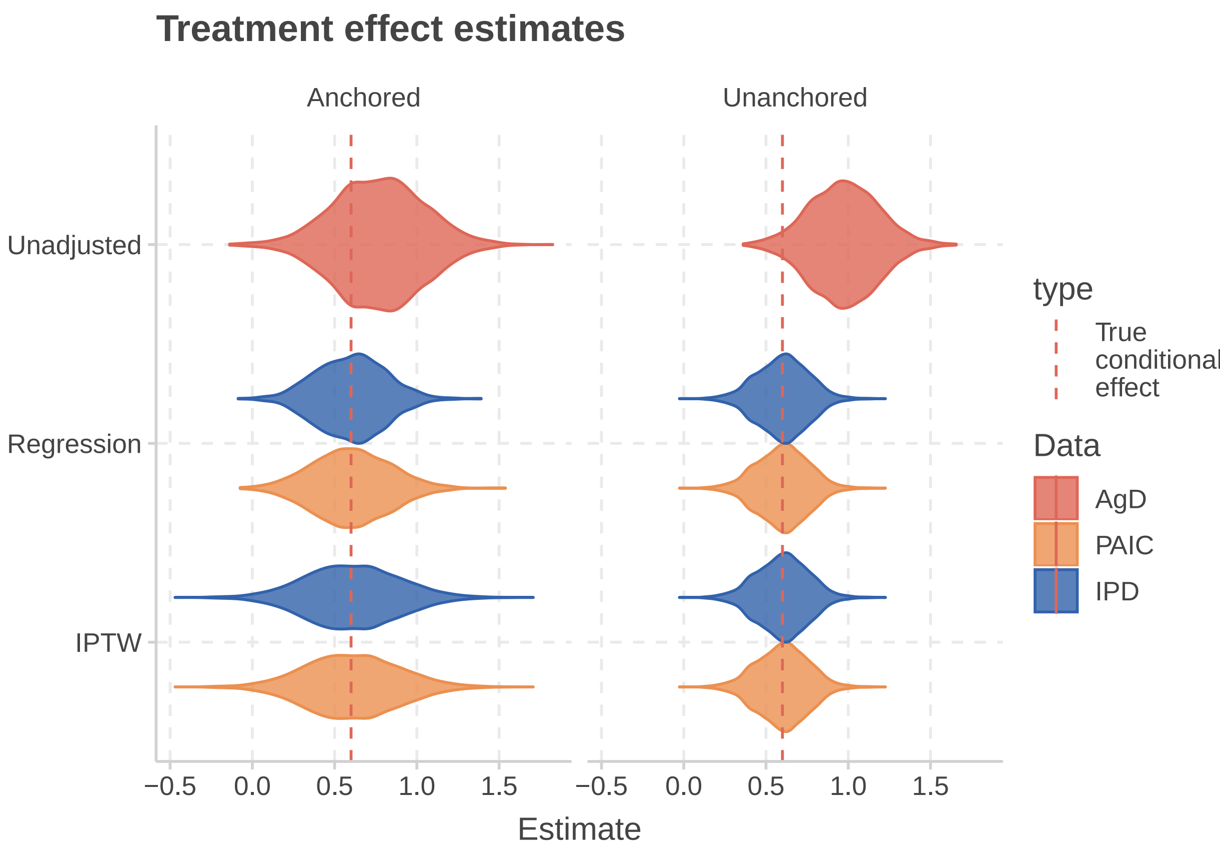
```
library(data.table)
N_pop <- 10^6

pop_init <- data.table(
  id = 1:N_pop,
  X1 = rbinom(N_pop, 1, prop_X1)
)

trial_assignement_prob <- function(df, trial_assignment_model, deviates = FALSE) {
  predicted <- with(df, eval(trial_assignment_model))
  # variance of a logistic function is set ((s^2\pi^2)/3), with s=1 when using normal law -->
  # making an assumption about bias and residual heterogeneity
  return(plogis(predicted))
}

predict_outcome <- function(df, outcome_model, deviates = FALSE) {
  predicted <- with(df, eval(outcome_model))
  if (deviates) return(predicted + rnorm(n = length(predicted), mean = 0, sd = 1)) else return(predicte
}

pop_init$prob_AC <- trial_assignement_prob(pop_init, trial_assignment_model_AC)
pop_init$prob_BC <- trial_assignement_prob(pop_init, trial_assignment_model_BC)


pop_init$Y_theo_A <- pop_init[, c("A", "B", "C") := .(1L, 0L, 0L)] |>
  predict_outcome(outcome_model)
pop_init$Y_theo_B <- pop_init[, c("A", "B", "C") := .(0L, 1L, 0L)] |>
```

```r
  predict_outcome(outcome_model)
pop_init$Y_theo_C <- pop_init[, c("A", "B", "C") := .(0L, 0L, 1L)] |>
  predict_outcome(outcome_model)
pop_init[, c("A", "B", "C") := NULL]

pop_init$prob_AC <- trial_assignement_prob(pop_init, trial_assignment_model_AC)
pop_init$prob_BC <- trial_assignement_prob(pop_init, trial_assignment_model_BC)

ttt_names <- c("A", "B", "C")
cols_obs <- paste0("Y_obs_", ttt_names)
cols_theo <- paste0("Y_theo_", ttt_names)
# Assigning "observed" values for all the individual in the population, ie theoretical outcome + gaussi
pop_init[, eval(bquote(cols_obs)) := lapply(.SD, \(x) x + rnorm(n = length(x))), .SD = cols_theo]
```

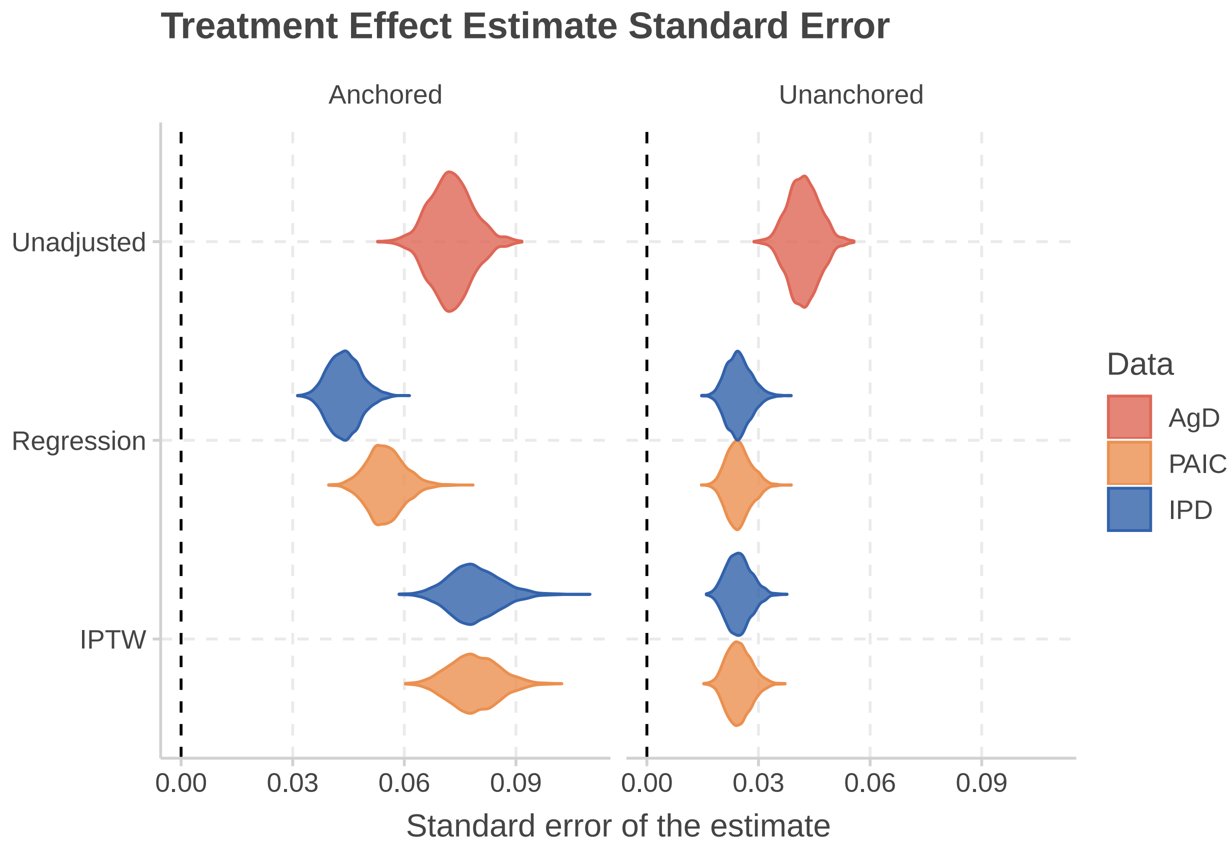
```r
trial_AC <- pop_init[
  sample(pop_init$id, N_RCT, replace = FALSE, prob = pop_init$prob_AC),
  .(id, ttt = factor(rep_len(c("A", "C"), length.out = N_RCT), levels = c("C", "A", "B")))][
    pop_init_long[,c("id", "X1", "ttt", "prob_AC", "theo", "obs")], on = .(id, ttt), nomatch = NULL
  ]
trial_BC <- pop_init[
  sample(pop_init$id, N_RCT, replace = FALSE, prob = pop_init$prob_BC),
  .(id, ttt = factor(rep_len(c("B", "C"), length.out = N_RCT), levels = c("C","A", "B")))][
    pop_init_long[,c("id", "X1", "ttt", "prob_BC", "theo", "obs")], on = .(id, ttt), nomatch = NULL
  ]
```

Simulation results

Distribution of the estimate

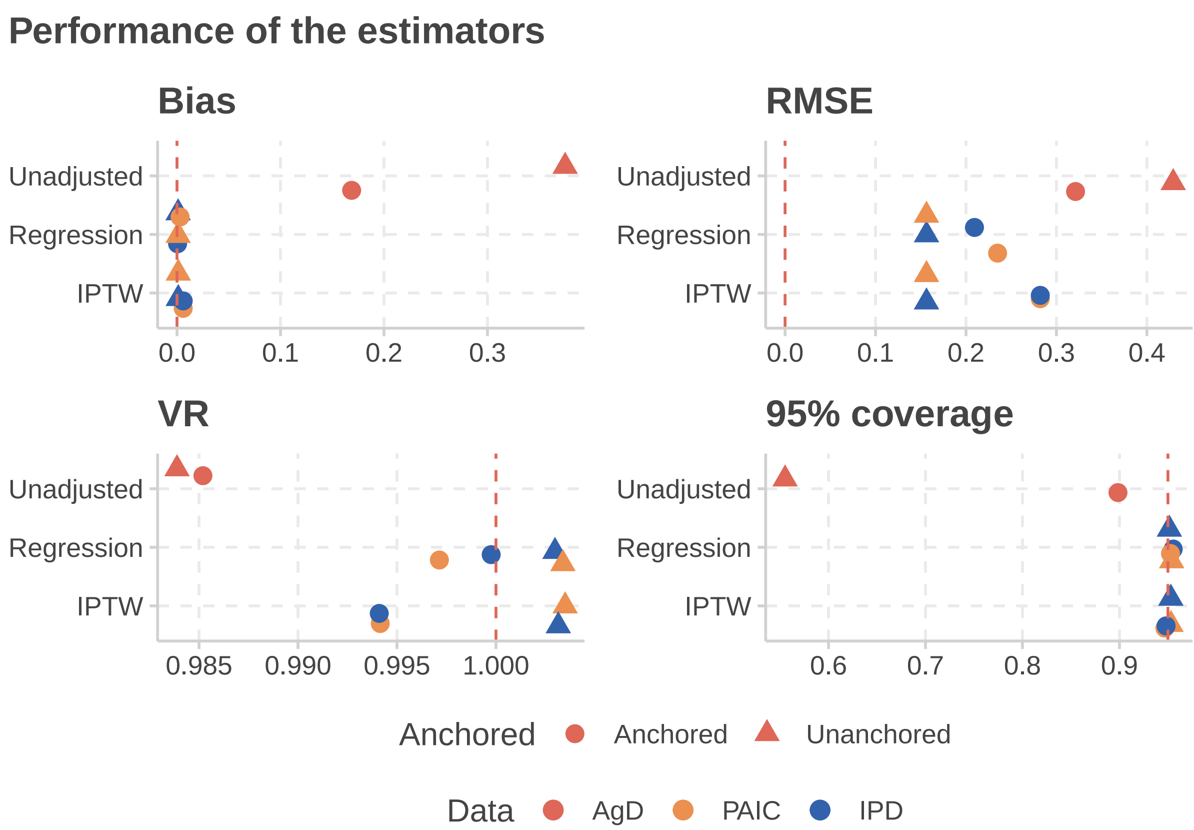


Standard error of the estimate



Performance of the estimators

indicator	Adjustment	Model	Anchored	Data	values
Bias	Unadjusted	Unadjusted	Unanchored	AgD	0.375
Bias	Unadjusted	Unadjusted	Anchored	AgD	0.169
Bias	Regression	STC	Unanchored	PAIC	0.001
Bias	Regression	STC	Anchored	PAIC	0.003
Bias	Regression	GLM	Unanchored	IPD	0.001
Bias	Regression	GLM	Anchored	IPD	0.001
Bias	IPTW	MAIC	Unanchored	PAIC	0.001
Bias	IPTW	MAIC	Anchored	PAIC	0.006
Bias	IPTW	ML	Unanchored	IPD	0.001
Bias	IPTW	ML	Anchored	IPD	0.006
Cov 95	Unadjusted	Unadjusted	Unanchored	AgD	0.555
Cov 95	Unadjusted	Unadjusted	Anchored	AgD	0.898
Cov 95	Regression	STC	Unanchored	PAIC	0.954
Cov 95	Regression	STC	Anchored	PAIC	0.953
Cov 95	Regression	GLM	Unanchored	IPD	0.952
Cov 95	Regression	GLM	Anchored	IPD	0.956
Cov 95	IPTW	MAIC	Unanchored	PAIC	0.953
Cov 95	IPTW	MAIC	Anchored	PAIC	0.947
Cov 95	IPTW	ML	Unanchored	IPD	0.953
Cov 95	IPTW	ML	Anchored	IPD	0.948
RMSE	Unadjusted	Unadjusted	Unanchored	AgD	0.429
RMSE	Unadjusted	Unadjusted	Anchored	AgD	0.321
RMSE	Regression	STC	Unanchored	PAIC	0.156
RMSE	Regression	STC	Anchored	PAIC	0.235
RMSE	Regression	GLM	Unanchored	IPD	0.156
RMSE	Regression	GLM	Anchored	IPD	0.209
RMSE	IPTW	MAIC	Unanchored	PAIC	0.156
RMSE	IPTW	MAIC	Anchored	PAIC	0.282
RMSE	IPTW	ML	Unanchored	IPD	0.156
RMSE	IPTW	ML	Anchored	IPD	0.282
VR	Unadjusted	Unadjusted	Unanchored	AgD	0.984
VR	Unadjusted	Unadjusted	Anchored	AgD	0.985
VR	Regression	STC	Unanchored	PAIC	1.003
VR	Regression	STC	Anchored	PAIC	0.997
VR	Regression	GLM	Unanchored	IPD	1.003
VR	Regression	GLM	Anchored	IPD	1.000
VR	IPTW	MAIC	Unanchored	PAIC	1.003
VR	IPTW	MAIC	Anchored	PAIC	0.994
VR	IPTW	ML	Unanchored	IPD	1.003
VR	IPTW	ML	Anchored	IPD	0.994



Performance of the estimators
Bias
RMSE
VR
95% coverage
Unadjusted
Regression
IPTW
0.0
0.1
0.2
0.3
0.4
0.985
0.990
0.995
1.000
0.6
0.7
0.8
0.9
Anchored
Anchored
Unanchored
Data
AgD
PAIC
IPD